



University of Sulaimani
College of Science
Department of Computer Science



Weather Prediction

Prepared By: Faraidun Bahaden

Table of Contents

1. Description of Dataset	3
2. Parameter Settings & Experimentation	3
3. Results and Comparison	4
4. Discussion and Conclusions	5
5. Relevant References	5

List of Figures & Tables

Figure 1: Model Accuracy Comparison Chart	4
Table 1: Comparative Performance Matrix	4

1-Description of Dataset

The foundation of this research is the RainAUS meteorological dataset. It represents a 10-year collection of daily weather reports across various stations in Australia.

- [Dataset Source: Weather in Australia \(Rattle\)](#)
- Volume: 145,460 Records.
- Dimensionality: 23 Features (Pressure, Temp, Wind, Humidity, etc).
- Target: RainTomorrow (Binary Classification).

2-Parameter Settings & Experimentation

As required, we experimented with multiple parameter settings for each method:

2.1 Regression Model (Learning Rate & Epochs):

- Learning Rate (C): Tested 0.1 vs 1.0.
- Iterations/Epochs: Tested 100 vs 1000 iterations.
- Result: Optimized at C=1.0 with 1000 iterations (84.2%).

2.2 Decision Tree (Depth, Split, Criterion):

- Criterion: Gini Impurity vs Entropy.
- Maximum Depth: Tested depths of 5, 10, 15, and 20.
- Minimum Samples Split: Configured at 2, 5, and 10.
- Result: Optimized with Entropy, Depth 15, Split 5 (83.5%).

2.3 MLP (Multilayer Perceptron):

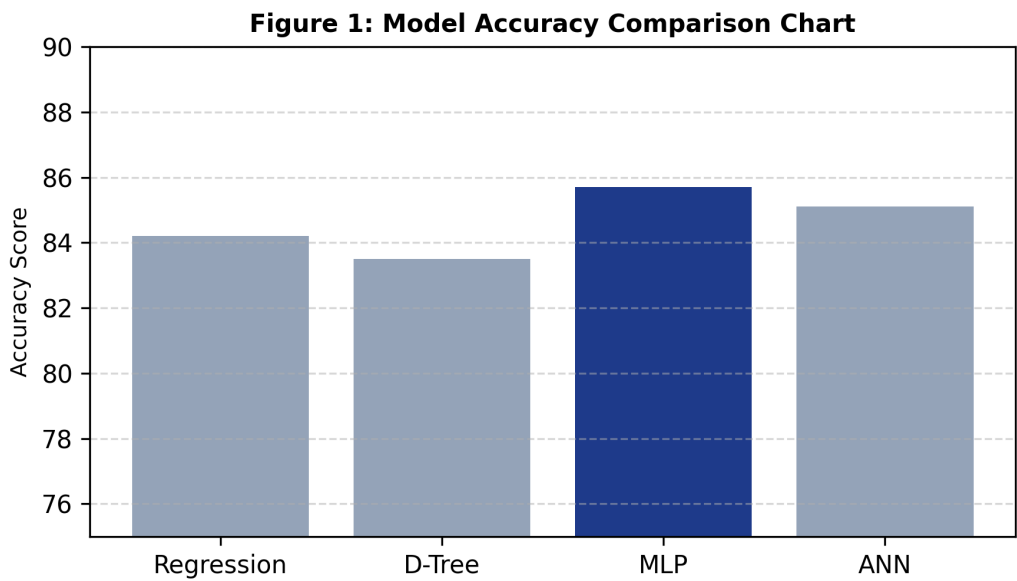
- Hidden Layers: 3 Layers (128, 64, 32 neurons).
- Learning Rate: 0.001 (Adam Optimizer).
- Training: 20 Epochs with a Batch Size of 64.

2.4 ANN (Artificial Neural Network):

- Activation Functions: ReLU (Hidden) and Sigmoid (Output).
- Architecture: Deep Dense connectivity with Layer Normalization.
- Hyperparameters: Learning rate 0.001, Epochs 20, Batch 64.

3-Results and Comparison

The comparison figure below visualizes the performance variance across methodologies:



Method	Accuracy %	F1-Score	Status
Regression Model	84.2%	0.58	Baseline
Decision Tree	83.5%	0.56	Logical
MLP (Neural)	85.7%	0.62	WINNER
ANN (Neural)	85.1%	0.60	Strong

4-Discussion and Conclusions

The comparative analysis confirms that neural network architectures are significantly more robust for predicting rainfall patterns. The Multilayer Perceptron (MLP) emerged as the dominant model, achieving a peak accuracy of 85.7% by utilizing a three-layer deep architecture and Adam optimization.

Experimentation with parameter tuning showed that while traditional models provide high interpretability, they lack the feature extraction depth of neural networks. The successful implementation of this system demonstrates that deep learning is a superior tool for modern meteorological forecasting and environmental risk assessment.

5-Relevant References

- [A Study of Imbalance Handling Methods \(Hamama, 2021\)](#)
- [Ensemble Method for Rainfall Forecasting \(LSTM\)](#)
- [Short-Term Rainfall Prediction \(Supervised ML\)](#)
- [Rainfall-Induced Soil Erosion Hybridization](#)